6.2.2 Amino acids, amides and chirality

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
Reactions of amino acids		
(a)	the general formula for an α-amino acid as $RCH(NH_2)COOH$ and the following reactions of amino acids: **(i)** reaction of the carboxylic acid group with alkalis and in the formation of esters **(see also 6.1.3 c)** **(ii)** reaction of the amine group with acids	
Amides		
(b)	structures of primary and secondary amides **(see also 6.1.3 f, 6.2.3 a–b)**	
Chirality		
(c)	optical isomerism (an example of stereoisomerism, in terms of non-superimposable mirror images about a chiral centre) **(see also 4.1.3 c–d)**	*M4.2, M4.3* Learners should be able to draw 3-D diagrams to illustrate stereoisomerism. HSW1,8
(d)	identification of chiral centres in a molecule of any organic compound.	*M4.2, M4.3*